

Comparative genomic analysis of mitogen activated protein kinase gene family in grapevine

Tae Kyung Hyun · Ju-Sung Kim · Suk-Yoon Kwon · Sun-Hyung Kim

Received: 1 February 2010 / Accepted: 24 February 2010 / Published online: 30 June 2010
© The Genetics Society of Korea and Springer 2010

Abstract

Mitogen activated protein kinases (MAPKs) are important proteins involved in the signal transduction of extracellular information to intracellular targets, and play a crucial role in the response to biotic and abiotic stresses. Although *Arabidopsis* MAPKs are used for identification of the putative MAPKs in higher plants, no grapevine MAPK gene nomenclature has yet been appeared in the literature. In this study, we have identified 12 members of grapevine MAPK gene (VvMPK) family via *In-silico* analysis of current grapevine genome database. The structural comparison of 12 VvMPKs through the analysis of chromosome locations, sequence annotation and paralogous gene pair indicated that VvMPKs have evolved by segmental duplication, rather than by tandem amplification. Although further functional analysis of VvMPKs through *in vivo* and *in vivo* experiments will be required, our study provides the basis for future research on the diverse signaling pathways mediated by MAPKs in grapevine.

Keywords Duplication; Grapevine; *In-silico* analysis; MAPK; Phylogenetic analysis

Introduction

Protein phosphorylation via protein kinases is the most common post-translational modification and one of the major mechanisms in controlling intracellular response to extracellular information (Jonak et al., 1999). In eukaryotic cells, it has been demonstrated that mitogen activated protein kinase (MAPK) cascades, one of the largest and most important categories of protein kinases, play an important role as universal modules of signal transduction (MAPK group, 2002; Shi et al., 2009). MAPKs form terminal components of the prototypical sequential cascades, and are activated by MAPK kinases (MAPKKs) via dual phosphorylating Thr and Tyr residues within TEY or TDY motif located in the activation loop (T-loop) between kinase subdomain VII and VIII (Kyiakis and Avruch, 1996; Widmann et al., 1999; Davis, 2000; Mishra et al., 2006).

In plants, MAPK cascades are regulated with various physiological, developmental and hormonal responses. A number of studies with molecular and biochemical approaches have revealed that the activation of MAPKs is not only involved in cell wall biosynthesis, cell growth, cell differentiation and hormone signaling, but also in stress response signaling such as pathogen infection, wounding, low temperature, drought, high salinity, and reactive oxygen species (Tena et al., 2001; Mishra et al., 2006; Zhang et al., 2006). Since genes encoding 20 MAPKs, 10 MAPKKs and more than 60 MAPKKKs were identified in the *Arabidopsis thaliana* genome (MAPK group, 2002), several dozens of MAPKs have been identified and isolated from tobacco (Seo et al., 1995; Zhang and Klessig, 1997; Romeis et al., 1999; Gomi et al., 2005), tomato (Holley et al., 2003; Mayrose et al., 2004), rice (Fu et al., 2002; Song and Goodman, 2002; Xiong and Yang, 2003; Liu and Xue, 2007), alfalfa (Jonak et al., 1996; Bogre et al., 1997), maize (Lalle et al., 2005), oat (Huttly and Phillips, 1995), wheat (Takezawa, 1999), barley (Knetsch et al., 1996), cotton (Wang et al., 2007), *petunia* (Decroocq-Ferrant et al., 1995) and poplar (Hamel et

T. K. Hyun
Institute for Plant Sciences, Karl Franzens-University of Graz,
Schubertstr. 51, A-8010 Graz, Austria

J.-S. Kim
Oriental Bio-herb Research Institute, Kangwon National University,
Chuncheon, 200-701 Korea

S.-Y. Kwon
Plant system Engineering Research Center, Korea Research
Institute of Bioscience and Biotechnology (KRIBB), Oun-dong
52, Yuseong-gu, Daejeon 305-806, Korea

S.-H. Kim (✉)
Department of Environmental Horticulture, The University of Seoul,
Jeonmang-dong 90, Dongdaemun-gu, Seoul 130-743, Korea
e-mail: mongdal@uos.ac.kr

al., 2006; Nicole et al., 2006). According to amino acid sequences, MAPKs share highly conserved region over the entire lengths with highest similarity in the eleven domains that are necessary for the catalytic function of serine/threonine protein kinases (MAPK group, 2002; Hamel et al., 2006).

Grapevine (*Vitis vinifera*), one of the most important economic crops, is a diploid plant with 19 chromosomes. The annotated genome sequences of grapevine have shown that a highly homozygous Pinot noir (PN40024) genome contains 30,434 genes in 487 Mb (The French–Italian Public Consortium for Grapevine Genome Characterization, 2007), and a highly heterozygous Pinot noir genotype (ENTAV 115) is predicted to have 29,585 genes within 531 Mb (Velasco et al., 2007). Although some supercontigs are still being correctly assembled and oriented in each chromosome (Matus et al., 2008), the completion of the grapevine genome has made it possible to identify gene family through the analysis of sequence similarity with *Arabidopsis* genome. In fact, terpene synthase gene family (The French–Italian Public Consortium for Grapevine Genome Characterization, 2007), MYB R2R3 subfamily (Matus et al., 2008) and invertase (Nonis et al., 2008) have been identified by *In-silico* analysis based on *Arabidopsis* genome. Recently, putative rice and poplar MAPKs, which are placed into four subgroups that are identified earlier in *Arabidopsis* based on the sequence similarity, have been revealed through *in-silico* analysis of current rice and poplar genome databases (Hamel et al., 2006; Nicole et al., 2006; Reyna and Yang, 2006; Liu and Xue, 2007). Although no nomenclature of grapevine MAPK genes has yet been appeared in the literature, these findings indicate the possibility of identification of grapevine MAPKs using *in-silico* analysis.

In this study, we conducted a genome-wide analysis of MAPK genes, and identified 12 putative MAPK genes by means of *in-silico* analysis of the grapevine genome sequence. Our genomic and bioinformatics analysis will provide the founda-

tion for further functional analysis of the MAPK signaling network in grapevine.

Materials and Methods

Databasesearch of MAPKs and sequence analysis

To identify MAPK genes from grapevine, grapevine genomic database (<http://www.genoscope.cns.fr/externe/GenomeBrowser/Vitis/>) was searched using 20 *Arabidopsis* MAPK protein sequences as direct queries with BLAT-search. In addition, the schematic representation of chromosomal positions and intron-harboring MAPK genes were also obtained from grapevine genomic database. Expressed sequence tags (ESTs) were identified by BLASTn of grape EST database (<http://cgf.ucdavis.edu/>), using PN40024 gene models as query sequences.

Conserved domain analysis and subcellular localization prediction

To analysis MAPK domain (PS01351), ATP-binding domain (PS00107), serine/threonine protein kinase active site (PS00108) and protein kinase domain (PS50011), motif scanning in 12 computationally predicted VvMPKs (*Vitis vinifera* MAPKs) was done by PROSITE scanning (<http://www.expasy.ch/prosite/>). Subcellular localization of each of VvMPKs was analyzed using CELLO v2.5 server (<http://cello.life.nctu.edu.tw/>).

Multiple-sequence alignment and construction of phylogenetic tree

Multiple-sequence alignment of computationally predicted VvMPK sequences was performed using ClustalW method (http://bbc.botany.utoronto.ca/ntools/cgi-bin/ntools_multi-align_w_mvview.cgi). The phylogenetic analysis was per-

Table 1. Nomenclature for MAPKs in grapevine

MAPK	Gene model name	Locus identifier	Genome location (5'-3')	AA length	Predicted subcellular localization
VvMPK1	GSVIVT00036900001	Vv4g1186	Chr 4: 13827460 - 13830742	368	nucleus
VvMPK2	GSVIVT00014559001	Vv18g1138	Chr 18: 10986583 - 10989612	371	nucleus
VvMPK3	GSVIVT00024457001	Vv6g0416	Chr 6: 4410189 - 4413681	375	cytoplasm
VvMPK4	GSVIVT00026919001	Vv15rg0220	Chr 15_random: 3106102-3111467	375	nucleus
VvMPK6	GSVIVT00034702001	Vv5g1656	Chr 5: 22671286 - 22692161	398	cytoplasm
VvMPK7	GSVIVT00017805001	Vv17g0259	Chr 17: 2372937 - 2382079	586	nucleus
VvMPK8	GSVIVT00030495001	Vv1g0504	Chr 1: 4545972 - 4551598	516	nucleus
VvMPK9	GSVIVT00024447001	Vv6g0426	Chr 6: 4557779 - 4561913	371	cytoplasm
VvMPK10	GSVIVT00019886001	Vv5g0445	Chr 5: 4111181 - 4119547	567	nucleus
VvMPK11	GSVIVT00027929001	Vv19g0025	Chr19: 212326 - 234450	623	nucleus
VvMPK12	GSVIVT00018914001	Vv12g0040	Chr 12: 550250 - 558226	610	cytoplasm